

Did You Do Well? Real-Time Personalized Feedback on Catheter Placement in Augmented Reality-assisted Neurosurgical Training

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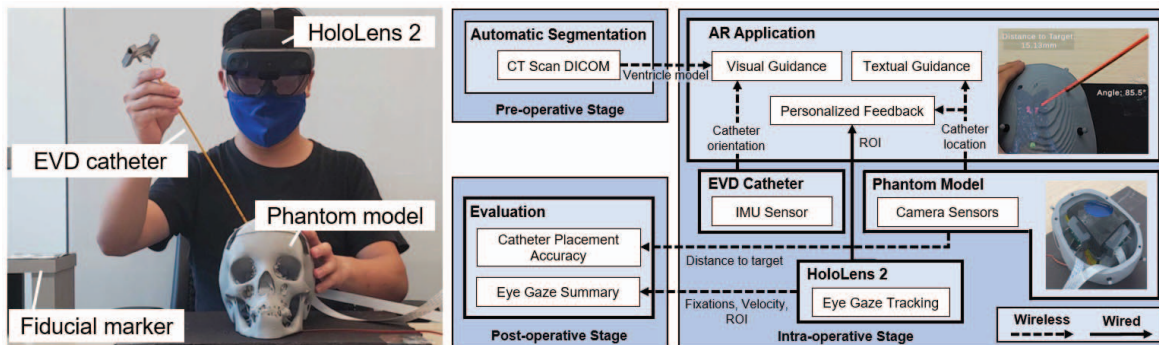


Figure 1: The hardware setup of our AR-assisted EVD system (left) and the overall architecture in three different stages: pre-operative, intra-operative, and post-operative (right).

ABSTRACT

External ventricular drain (EVD) is a common neurosurgical procedure often performed by junior trainees to relieve pressure buildup inside the brain by inserting a catheter to drain fluid. Recent work shows that the integration of Augmented Reality (AR) improves catheter placement accuracy by visualizing the brain ventricular anatomy and other guidance as holograms. However, trainees often require feedback on their overall performance. Therefore, we demonstrate an AR-assisted neurosurgical training tool that provides real-time personalized feedback to trainees based on their manipulation of the surgical environment and their eye gaze patterns. Furthermore, we developed a segmentation tool that automatically segments and visualizes brain ventricular models in AR. We showcase how real-time AR feedback can assist trainees in improving their EVD performances over various shapes of brain ventricular anatomies that pose varying levels of challenges.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed / augmented reality; Human-centered computing—Human computer interaction (HCI)—Interaction devices—Displays and imagers; Applied computing—Life and medical sciences—Health care information systems;

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1 INTRODUCTION

The integration of Augmented Reality (AR) in neurosurgical applications has shown the potential to assist surgeons through the visualization of patient's anatomy [2]. External ventricular drain (EVD) is a prime example that benefits from the integration of AR by providing intra-operative guidance to surgeons. EVD is performed on patients to relieve pressure buildup by placing a catheter into the brain ventricle system, which removes cerebrospinal fluid. Freehand insertion is a traditional approach to placing the EVD catheter without any aides by relying on external anatomical landmarks, often performed by junior trainees. However, they are more likely to miss the target, the foramen of Monro, and make multiple passes of the catheter, increasing the risk of complications such as hemorrhage or infections [5]. While prior work in AR-assisted EVD placement [1, 3, 4] provided guidance about catheter trajectory, trainees often require individual feedback about their overall performance such as their tool handling or attention patterns to learn and improve their behaviors for better performance.

Hence, in this demo, we present an AR-assisted neurosurgical training tool for EVD, as shown in Fig. 1a, that provides personalized feedback about the trainee's performance in response to their behavior. We developed a graphical user interface (GUI) on an edge server to segment brain ventricular models from CT images in Digital Imaging and Communications in Medicine (DICOM) format and to visualize a summary of the trainee's performance upon completion of the procedure. The AR app visualizes the ventricular model and target holograms and provides real-time guidance of the distance-to-target, catheter angle, and optimal trajectory. Additionally, the app provides personalized feedback that is dynamically-generated by changes in the trainee's eye gaze patterns and the trainee's manipulation of the catheter. This guidance helps junior trainees such as medical students or residents in estimating the target point and improving their catheter placement accuracy. Our system allows them to perform AR-assisted EVD trials on various brain ventricu-

lar anatomies that pose different levels of challenges and provides personalized feedback on their catheter placements.

2 SYSTEM DESIGN

Our AR-assisted neurosurgical training tool for EVD comes in three stages. In the pre-operative stage, the brain ventricular anatomy is segmented on the edge server and sent over to Microsoft HoloLens 2 via a wireless network for visualization. In the intra-operative stage, the segmented brain ventricle is registered to the phantom model via fiducial marker detection and the system provides multiple forms of real-time AR feedback personalized to the trainee's performance. Finally, in the post-operative stage, the summary of participants' performance and eye gaze data are provided.

Segmentation of brain ventricles: During the pre-operative stage, the system automatically segments the brain ventricles via a thresholding-based approach on the edge server. Our automatic segmentation segments the brain ventricle using DICOM image inputs and visualizes the segmented ventricular model as an AR hologram. A portion of our GUI allows the user to select parameters including the upper threshold, iterations of erosions and dilations, and the z-axis coordinate of nasion. Based on the parameter settings, the segmented brain ventricle is visualized on the GUI (Fig. 2). A mesh representation of the model is constructed using the marching cubes algorithm by connecting the ventricle vertices to create the ventricle faces. These data are then sent from the edge server to HoloLens 2 via a wireless network.

Sensing-integrated phantom: We designed a custom brain mold with two camera sensors attached to adjacent sides of the mold capturing image frames in real-time within the phantom skull, as shown in Fig. 1b. We use Arducam wide-angle OV5647 camera sensors that support a frame rate of up to 15fps at a resolution of 2592x1944. The camera sensors are equipped with M12 lenses that capture wide-angle images within the brain mold to ensure the catheter tip is visible on the captured images in all trajectories. The 3D locations of the catheter tip and target are triangulated, and the orientation of the catheter is obtained from an inertial measurement unit sensor attached to the top of the catheter. This data is sent to the HoloLens 2 via a wireless network.

AR guidance: After receiving the brain ventricle data, the ventricular hologram is registered to the phantom model by detecting a fiducial marker. Once the trainee manipulates the catheter, the catheter hologram and text guidance are updated. The system provides catheter placement accuracy in real time by visualizing the distance from the catheter tip to the target and the catheter insertion angle as text in AR, as seen in Fig. 1c. The catheter hologram is visualized based on the catheter tip location and the catheter orientation obtained from sensors, and its color changes from red to green when the trainee aligns the catheter with the optimal trajectory. This trajectory is visualized as a dotted line between the tip and the target.

Personalized feedback: Additionally, the app presents several forms of personalized feedback to the user. The trainee's eye gaze patterns are analyzed and holograms that have not been focused on in the last 30 seconds flash until the trainee directs their attention to them. Additionally, if the 3D location of the catheter tip is detected outside of the acceptable bounds of the phantom model, a textual warning is issued. Finally, if the trainee performs more than four passes of catheter insertion within one trial, a textual warning is issued. Upon completion of the trial, a summary of the trainee's performance is visualized on the GUI (Fig. 2).

3 INTERACTIVE DEMONSTRATION

Our demonstration uses the GUI in Fig. 2 and the sensing-integrated phantom model in Fig. 1b. We use the Microsoft Surface Laptop 4 with Intel Core i7 and built-in GPU as the edge server. The demo allows the participants to develop an understanding of the AR feedback that provides benefits of visual and contextual feedback

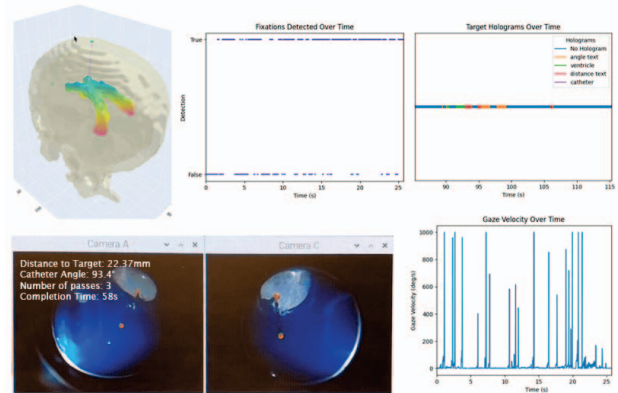


Figure 2: GUI dashboard showing participants' EVD performance.

in inserting an EVD catheter targeting the foramen of Monro of a phantom model. The participants wear the HoloLens 2 to experience the AR app and insert the catheter with AR feedback. A video of the demo is available online¹.

During the demo, the participants select a brain CT scan using the GUI and configure the parameters for the system to segment the brain ventricles automatically. The segmented brain ventricular anatomy data will be sent over to HoloLens 2 via a wireless network for visualization inside the AR app. Then, the participants can insert a catheter through an entry point on the right side of the phantom with AR guidance showing the distance to the target point, angle, and trajectory, as well as personalized feedback based on their eye gaze patterns and surgical performance. After the procedure is complete, the participants can view a summary of their performance on the GUI. The participants can envision how AR feedback can assist medical students and novice surgeons in improving catheter placement accuracy during EVD training.

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¹Link to a demo video: <https://youtu.be/7gYBICLm2jg>